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MILITARY SPECIFICATION

MEAT TENDERIZER, ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electric meat tenderizer for tenderizing steaks, and blending and knitting small pieces of meat into tenderized steaks.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CC-M-1807	-	Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-60	-	Tape: Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Natick Research and Development Command, ATTN: DRXNM-TDG, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

MILITARY

MIL-P-116	-	Preservation-Packaging, Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirement for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurements of

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL SANITATION FOUNDATION (NSF)

No. 8 - Commercial Powered Food Preparation Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, MI 48106.)

Evidence of [illegible] except on [illegible]

UNDERWRITERS LABORATORIES, INC. (UL)

UL 73 - Motor-Operated Appliances

(Application for copies of UL Standards should be addressed to the Underwriters Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Rd., Northbrook, IL 60662; 1285 Walt Whitman Road, Melville, L.I., NY 11746; 1655 Scott Blvd., Santa Clara, CA 95050; or 2602 Tampa East Blvd., Tampa, FL 33619.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W. Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room, 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The meat tenderizer shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

3.2 Codes and standards. The meat tenderizer shall comply with the applicable requirements of NSF Standard No. 8 and UL 73.

3.2.1 Compliance. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative, that the meat tenderizer conforms to the applicable requirements of UL 73 and NSF Standard No. 8.

3.2.1.1 UL. Acceptable evidence of meeting the requirements of UL 73 shall be the UL label or the UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the meat tenderizer has been tested and conforms to UL 73.

3.2.1.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finsihed meat tenderizer, or
(2) A certification for the meat tenderizer issued by NSF under their special one-time contract evaluation/certification service, or
(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the meat tenderizers are being procured, indicating that the meat tenderizers have been tested and conform to the applicable requirements of NSF Standard No. 8.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial meat tenderizers, provided the completed item complies with all provisions of this specification.

3.4 Design and construction. The meat tenderizer shall consist of a steel or aluminum base and frame or may be provided with legs. It shall essentially consist of a cutting group housed within a heavy gage metal, aluminum or high impact plastic housing, together with an electric motor driving mechanism and a lid covering the cutting group. The lid shall be fabricated from corrosion resistant metal, aluminum, or a transparent high impact plastic. The tenderizer shall have a safety switch to prevent the machine from operating when the lid is in the open position. The meat feed opening shall be not less than 7-1/2 inches (190 mm) long by not more than 1-1/4 inches (32 mm) wide. The meat tenderizer shall be provided with feet mounted on the base to prevent slipping. When legs are provided they shall be not less than 5 inches (127 mm) nor more than 8 inches (203 mm) in height and shall be adjustable.

3.4.1 Cutting group. The cutting group shall consist of roller type serrated cutting knives, tines, and guides assembled together as a complete unit. The cutting group shall be designed to enable easy removal as a unit.

3.4.1.1 Cutting knives. The cutting knives shall be fabricated of stainless steel. Knives shall be positively locked on the shaft to prevent slipping, and readily removable from the shaft for individual replacement. A minimum spacing of 5/32 inch (4 mm) and a maximum spacing of 7/32 inch (6 mm) between the cutting knives shall be maintained. The cutting knives shall have a hardness of 42 to 48 on the Rockwell "C" scale when tested as specified in 4.4.2.

3.4.2 Electrical characteristics. The meat tenderizer shall operate on a nominal 120 volt, 60 hertz, single phase power source, completely wired and equipped with a three conductor cord, 6 foot (1.8 m) minimum length, and three prong grounding plug.

3.4.2.1 Motor. The motor shall be alternating current (ac), continuous duty, capacitor start, induction type and shall be not less than 1/3 horsepower conforming to CC-M-1807. Motor shall be enclosed by the machine housing, except at the bottom, and shall be of the enclosure furnished regularly by the manufacturer on tenderizers, bearing listing approval by Underwriters Laboratories. The motor shall be equipped with an automatic thermal cut out device. When specified (see 6.2), the motor shall be 3 phase, squirrel cage, induction type.

3.4.2.2 Switch. The operating "on" and "off" switch shall be of the commercial type having a minimum rating of 1 horsepower, double pole to open both sides of the line. The switch shall be provided with a guard to prevent accidential operation of the switch.

3.4.2.3 Electromagnetic compatibility. When specified (see 6.2), the meat tenderizer shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461.

3.4.3 Navy shipboard installation. When specified (see 6.2), for Navy installation, the electrical characteristics shall be as specified in 3.4.2 or 120 volt direct current completely wired and equipped with a three conductor cord, 6 foot (1.8 m) minimum length, and a three prong grounding type plug.

3.4.3.1 Mounting. The meat tenderizer shall be provided with means for bolting to a table top.

3.4.4 Drive. The cutting knives shall be V-belt or gear driven or a combination of both. V-belts shall be a commercial grade, oil, water, and heat resistant. Gears shall have smooth teeth properly meshed and operate in a lubricant in a sealed housing.

3.4.5 Performance. The meat tenderizer shall be capable of processing flank steak approximately 7 inches (178 mm) diameter by 1 inch (25 mm) thick without vibration or objectionable noise, when tested as specified in 4.4.1.

3.4.6 Noise limits. The food mixing machines shall not exceed 85 decibels A-weighted at the normal operator position when tested as specified in 4.4.4.

3.4.7 Cleaning. The meat tenderizer shall be supplied with a cleaning tool to remove excess meat particles between the knives, or so designed that the strippers will clean excess meat particles from the cutting knives.

3.5 Finish. The finish shall be in conformance NSF Standard No. 8.

3.6 Identification marking. Identification marking shall be permanently and legibly marked directly on the item or on an aluminum, brass, or stainless steel plate firmly affixed to the item. Marking shall be stamped, embossed, engraved, or applied by photosensitive means. The plate thickness shall be not less than 0.012 inches (0.3 mm). The identification plate shall include the name, trade name or trademark, model or serial number of such known character as to be readily identifiable to the manufacturer.

3.7 Workmanship. All components and assemblies of the meat tenderizer shall be free from dirt and other extraneous materials; burrs, slivers; rough die, tool and grind marks; dents and cracks. Castings, moulded parts and stampings, if used, shall be free from sharp edges and corners.

3.7.1 Metal fabrication. Metal used in the fabrication of the meat tenderizer shall be free from kinks. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and of uniform size and shape.

3.7.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts or incomplete fusion. All scale and flux shall be removed from the finished weld area.

3.7.3 Threaded fasteners. Threaded fasteners and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked or stripped and shall be securely torqued. Rivets shall fill the holes completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, standards, and drawings unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 In-process inspection.

4.2.2.1 Hardness testing. The cutting knives shall be tested for hardness in accordance with 4.4.2. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects, expressed in terms of defects per hundred units. The sample unit shall be one cutting knife. The lot shall consist of all cutting knives of one type offered for acceptance at one time.

4.2.3 End item inspection. The lot shall consist of all meat tenderizers offered for inspection at one time. The sample unit shall be a completely fabricated meat tenderizer.

4.2.3.1 Visual examination. The meat tenderizer shall be examined for defects classified in table I. The inspection level shall be II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of visual defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturers standard product except for changes necessary to comply with specification requirements	X	
Finish	Not as specified	X	
	Rough or sharp edges, burr(s) slivers, or scratches	X	
Construction and workmanship general	Component missing	X	
	Any burr, sliver, sharp edge, rough die, tool, gouge, trim or grind marks	X	
	Not fabricated of material specified	X	
	Component split, cracked, or showing delamination	X	

TABLE I. Classification of visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship general (cont'd)	Any component fractured, buckled, bent, dented, punctured or malformed	X	
	Any component misplaced or not in alignment	X	
Fastening devices	Missing, cracked or stripped	X	
	One bolt, nut, screw or washer missing	X	
	Two or more bolts, nuts, screws or washer missing	X	
	Fastener loose		X
	Rivet holes not accurately punched	X	
	Rivet not peened properly		X
Welding	Not free from rust scale, paint, grease and other foreign matter	X	
	Scale and flux deposits not removed		X
	Weld not smooth and free of cracks, burn holes, undercuts or incomplete fusion		X
Castings, molded parts and stampings	Miscast or incomplete, not free of sand, fins, pits, sprues and blow-holes	X	
Design and construction	Not furnished with a safety switch	X	
	Legs not adjustable, when provided	X	
	Legs not provided with feet to prevent slipping	X	
	Cutting group not removable as a complete unit	X	
	Knives not positively locked on shaft	X	
	Power cord missing, not type specified, not equipped with 3-prong grounding plug	X	
	Motor not enclosed by machine housing, except at bottom	X	
	Motor not equipped with automatic thermal cut out device	X	
	Switch not double pole to open both sides of the line	X	
	Legs not provided with means for bolting to table top (Navy shipboard installation)	X	

TABLE I. Classification for visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Identification marking	Not permanently marked on the item or aluminum, brass or stainless steel plate	X	
	Missing, incomplete, illegible	X	

4.2.3.2 Dimensional examination. Examination shall be made of the end item to determine compliance with the dimensional requirements of this specification. Any dimension that is not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.2.3.3 End item testing. One representative production unit shall be tested in accordance with 4.4.1, 4.4.3 and 4.4.4, when applicable. Any nonconformance shall be cause for rejection of the unit.

4.2.4 Packaging inspection. An examination shall be made to determine whether the preservation-packaging, packing and marking complies with the Section 5 requirements. Defects shall be as listed in table II. The sample unit shall be one shipping container fully packaged, except that it need not be closed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

TABLE II. Examination of packaging

Examine	Defect
Marking (exterior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing, damaged or not as specified.
Preservation	Preservative improperly applied or missing.
Workmanship	Unit not sealed, protective barrier material, when applicable, not secured. Contents of container not immobilized, cushioned, block or braced. Inadequate application of sealed case liner in wood container. Bulged or distorted containers.
Content	Contents per container is more than required.

4.2.4.1 Examination of shipping containers. When shipping containers are required to comply with PPP-B-601 or PPP-B-636, examination for defects in the closure, waterproofing (for fiberboard containers), and reinforcing shall be in accordance with the appendix of the applicable container specification.

4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Test methods.

4.4.1 Performance test. The assembled meat tenderizer shall be connected to an electric power source having the same characteristics as specified and operated without load for 30 minutes. A fresh raw piece of meat having a temperature of 35 + 5°F (1.7 + 3°C), approximately 7 inches (178 mm) diameter by 1 inch (25 mm) thick, shall be processed to determine conformance with 3.4.5. While the meat tenderizer is operating the lid shall be raised causing the machine to stop. Any noncompliance with the requirements specified shall constitute failure of this test.

4.4.2 Hardness test. Cutting knives shall be tested for hardness in accordance with ASTM E 18 to determine compliance with 3.4.1.1. Any nonconformance to meet the hardness requirements specified shall constitute failure of the test.

4.4.3 Electromagnetic compatibility test. When electromagnetic compatibility is required, the production unit shall be tested by the contractor in accordance with test method CE03, and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or his private testing organization. The contractor shall furnish to the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Compatibility Test Report and requirements meet MIL-STD-461.

4.4.4 Noise test. The food mixing machine shall be tested in accordance with MIL-STD-1474 to determine conformance with 3.4.6. Any nonconformance shall constitute failure of this test.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. All exposed uncoated ferrous metal surfaces shall be cleaned process C-1 of MIL-P-116 and then thoroughly dried.

5.1.1.2 Application of preservative. All cleaned and dried surfaces, except food contacting surfaces, shall be coated with type P-2 preservative of MIL-P-116. All cleaned and dried surfaces which will be in contact with food shall be coated with type P-14 preservative of MIL-P-116.

5.1.1.3 Lubrication. All bearings and fittings requiring lubrication shall be lubricated to capacity with the manufacturer's recommended lubricant containing a rust inhibitor.

5.1.1.4 Sealing. Openings into the motor, switches and crevices into the unit, shall be sealed with tape conforming to type III or IV of PPP-T-60.

5.1.1.5 Protection of preservatives. Coated surfaces from which a preservative might be removed by contact with packaging material shall be covered, wrapped or bagged with barrier material conforming to type I or II, grade A, class 1 or 2 of MIL-B-121. The barrier material shall be secured with the tape specified in 5.1.1.4 or be sealed, when applicable.

5.1.1.6 Line cord. The line cord shall be looped and securely tied in place.

5.1.1.7 Securing. Parts of the unit that are free to move shall be secured in a fixed position to prevent movement or vibration while in transit.

5.1.1.8 Preservation of cleaning tool. The cleaning tool, when applicable, shall be cleaned and dried as specified in 5.1.1.1, coated as specified in 5.1.1.2, wrapped in barrier material as specified in 5.1.1.5 and be securely taped to the unit with the tape specified in 5.1.1.4.

5.1.2 Level C. Each meat tenderizer shall be preserved-packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A packing. One meat tenderizer, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC-L, grade V2s of PPP-B-636 or overseas type, style A or J, grade A or B, type 2 load of PPP-B-601. The inside of each fiberboard container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The contents of each container shall be cushioned, blocked or braced within the container to prevent movement while in transit. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636. Each wood container shall be waterproofed with a sealed case liner conforming to type I or II, grade C of MIL-L-10547 and be closed and reinforced in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. One meat tenderizer, preserved-packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. The contents of each container shall be cushioned, blocked or braced within the container to prevent movement while in transit. Each fiberboard container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.3 Level C packing. Each meat tenderizer, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as normally used by the contractor for retail distribution. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tenderizer covered by this specification is intended to blend and knit small pieces of meat into tenderized steaks.

6.2 Ordering data. Procurement documents should specify the following.

(a) Title, number and date of this specification.
(b) When 3 phase, squirrel cage induction type motor is required (see 3.4.2.1).
(c) When electromagnetic compatability is required (see 3.4.2.3).
(d) Electrical characteristics for Navy shipboard installation (see 3.4.3).
(e) Selection of the applicable levels of preservation-packaging and packing (see 5.1 and 5.2).
(f) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 7320-0635

Review activities:

Army - MD
Navy - SA

User activities:

Navy - MC, MS

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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